

Childcare Affordability in the United States

Data-Driven Insights from the National Database of
Childcare Prices

AYDEN TRUSTY
DSC640
7 MARCH 2026

Why This Matters

- ▶ Childcare affordability varies significantly across the United States. In many regions, childcare costs consume a large portion of household income, creating financial pressure for families.
- ▶ Understanding these cost differences is important because childcare affordability affects workforce participation, economic mobility, and family financial stability.
- ▶ This analysis explores national childcare cost trends, compares care types, and examines geographic differences in affordability.

Dataset Overview

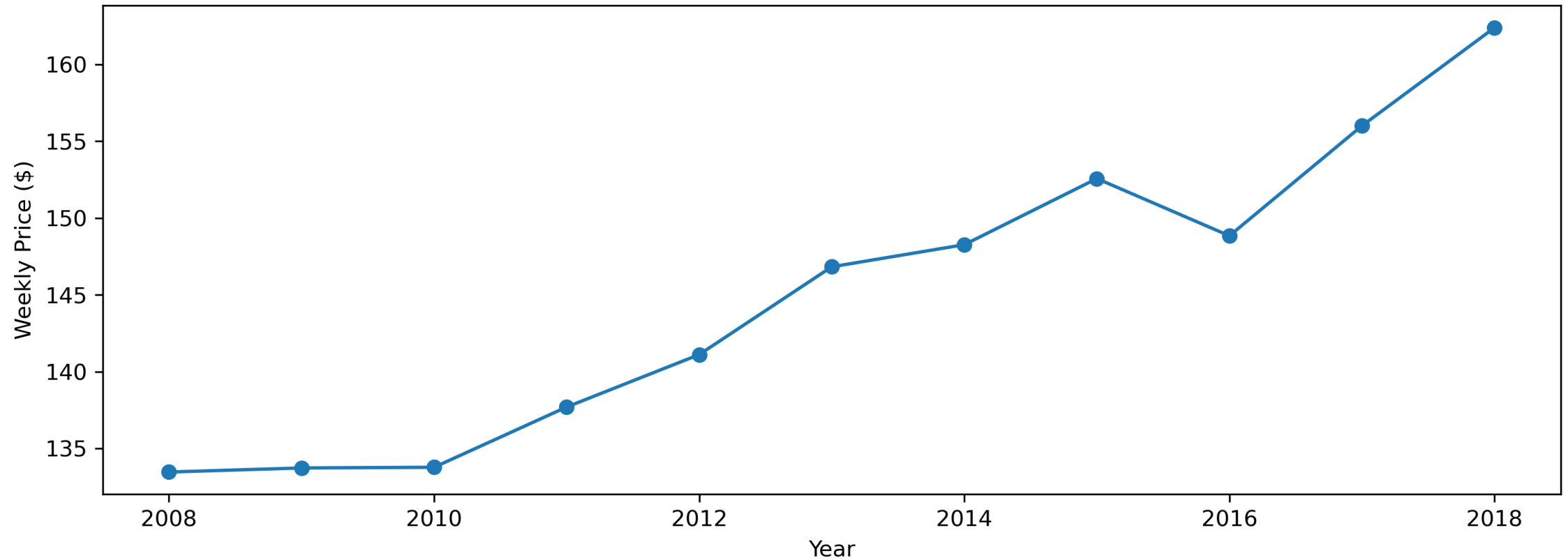
- ▶ The dataset used in this project comes from the **National Database of Childcare Prices**, which provides modeled childcare price estimates across the United States.
- ▶ The dataset includes:
 - ▶ County-level childcare price estimates
 - ▶ Data spanning **2008–2018**
 - ▶ Different childcare types (center-based vs family childcare)
 - ▶ Median household income data
- ▶ These variables allow for analysis of childcare costs relative to income levels.

Medium 1: Presentation Analysis

- ▶ The first medium used in this project is a presentation format designed to communicate the key findings clearly to a broad audience.
- ▶ The goal of this section is to highlight the most important insights from the data, including trends in childcare cost, differences between childcare types, and the overall affordability burden.
- ▶ These visuals focus on storytelling and making the analysis accessible to policymakers or stakeholders who may not have a technical background.

National Trend Over Time

Average Weekly Infant Center-Based Childcare Price Over Time

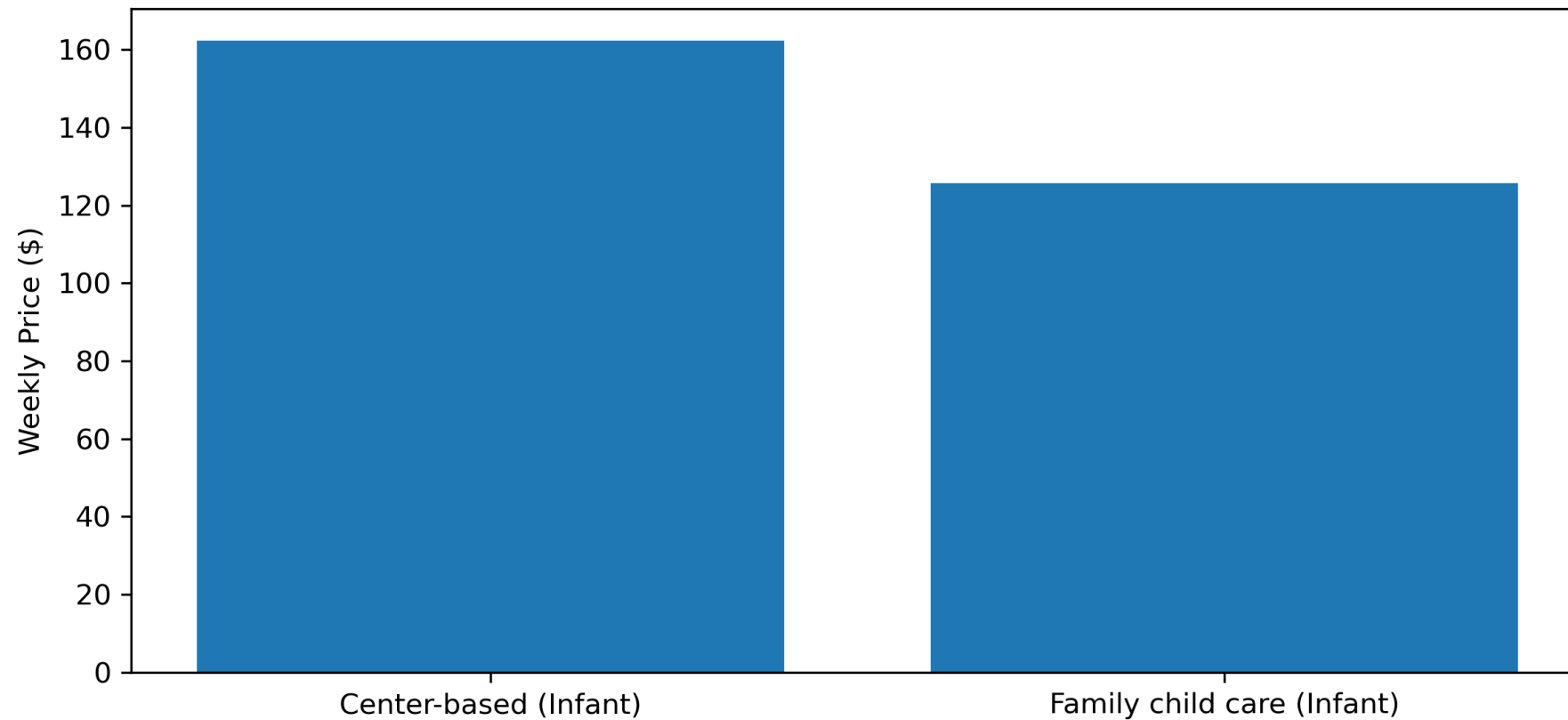


National Trend Over Time

- ▶ Infant center-based childcare costs have steadily increased between 2008 and 2018. This trend indicates that childcare expenses have grown over time, increasing financial pressure on families.
- ▶ Even modest yearly increases accumulate into substantial cost differences over time, particularly for households already managing tight budgets.

Care Type Price Difference

Center vs Family Child Care Infant Prices (2018 Average)



Care Type Price Difference

- ▶ Center-based infant childcare is significantly more expensive than family childcare on average.
- ▶ While family childcare may provide a more affordable option in some cases, the overall cost of infant care remains high relative to income levels in many regions.
- ▶ This difference in care type pricing can influence family childcare decisions.

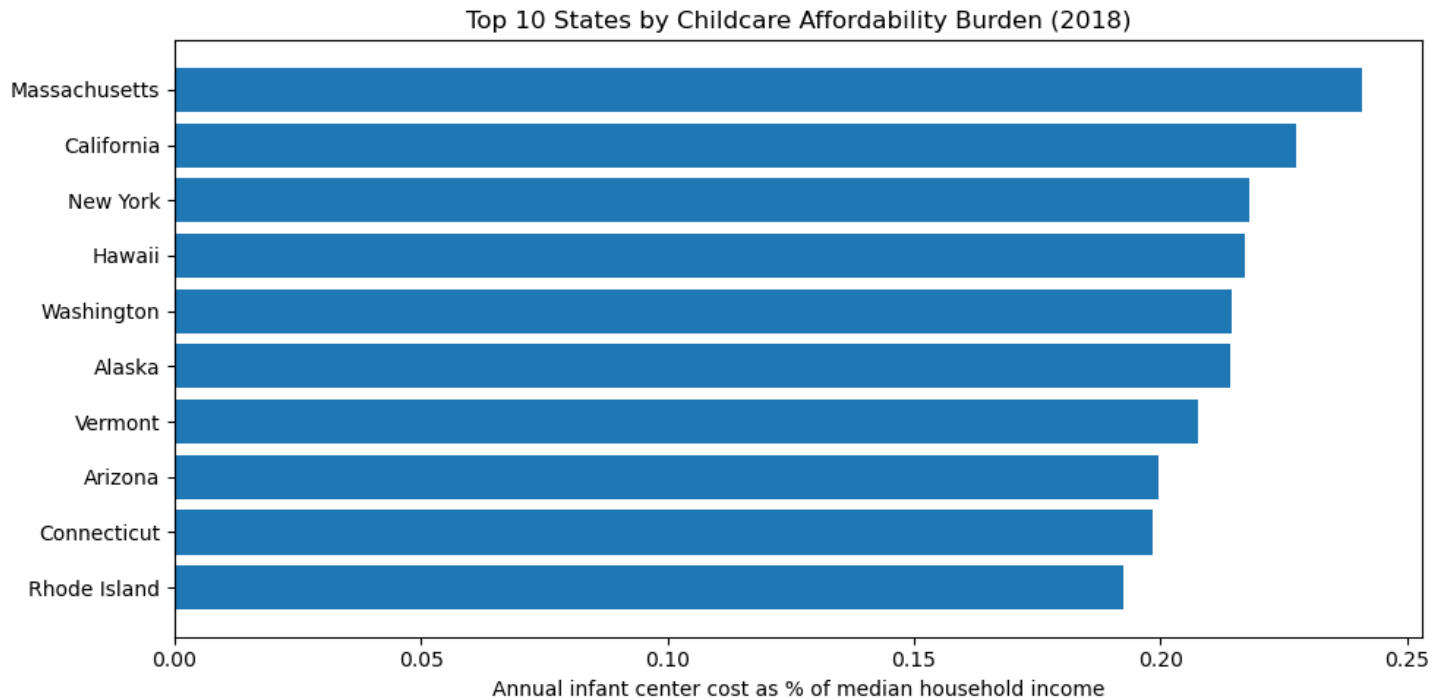
Medium 2: Analytical Notebook (Jupyter)

- ▶ The second medium used in this project is a Jupyter Notebook, which was used to perform the technical analysis behind the visuals.
- ▶ The notebook includes:
 - ▶ Loading and cleaning the dataset
 - ▶ Calculating annual childcare cost
 - ▶ Computing affordability burden
 - ▶ Generating supporting visualizations
- ▶ This medium provides transparency by showing how the data was processed and analyzed.

Affordability Burden Calculation

- ▶ Affordability burden was calculated by dividing annual infant childcare cost by median household income.
- ▶ Annual childcare cost was estimated by multiplying weekly childcare prices by 52 weeks.
- ▶ This calculation allows for comparison of childcare costs relative to income rather than simply comparing raw price values.

States with Highest Affordability Burden



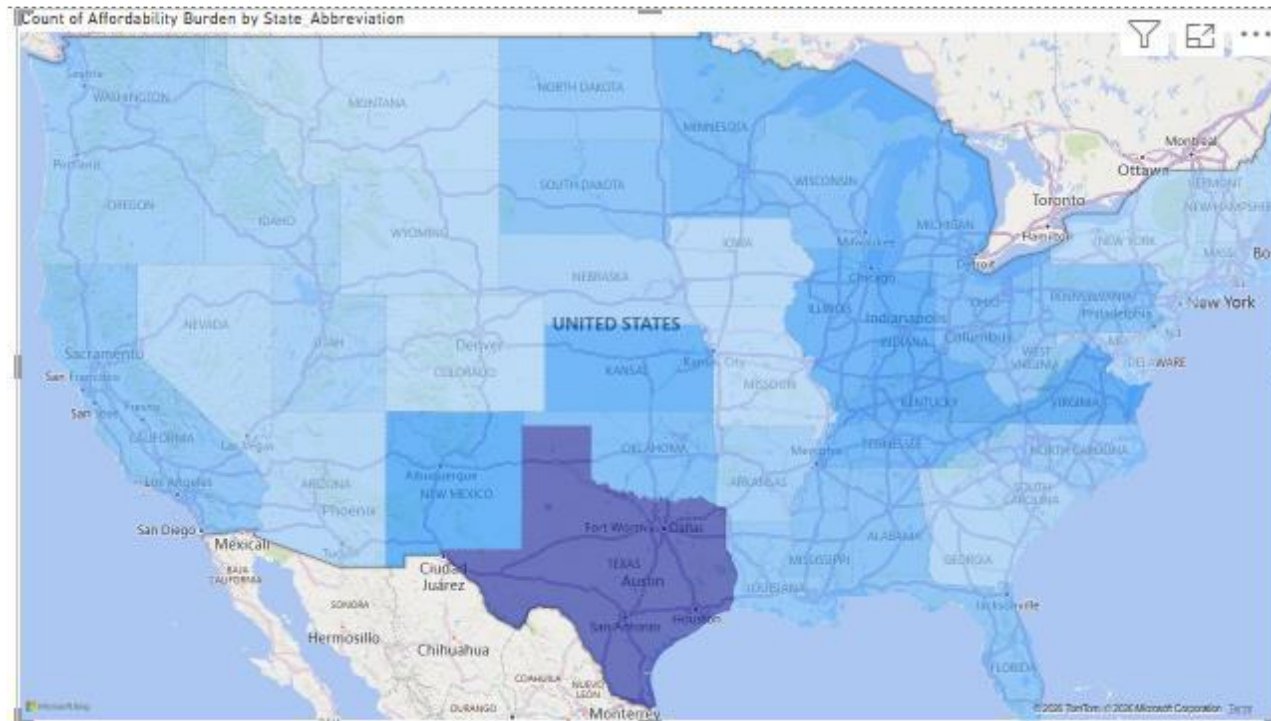
States with Highest Affordability Burden

- ▶ Several states experience particularly high childcare affordability burdens, where infant childcare costs represent a large percentage of median household income.
- ▶ This demonstrates that affordability challenges are not evenly distributed and depend heavily on regional economic conditions.

Medium 3: Interactive Dashboard

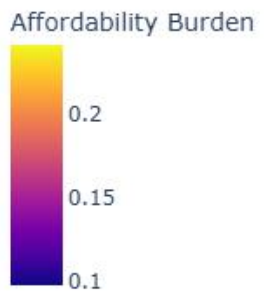
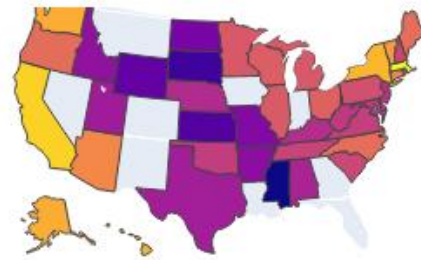
- ▶ The third medium used in this project is a Power BI dashboard, which allows users to explore geographic differences in childcare affordability.
- ▶ The dashboard visualizes affordability burden across states using a map-based visualization.
- ▶ While the submitted version includes screenshots of the dashboard, the original version allows users to interact with the data and examine patterns more closely.

100



Geographic Distribution of Childcare Affordability Burden

Childcare Affordability Burden by State (2018)



Geographic Distribution of Childcare Affordability Burden

- ▶ This heatmap illustrates the distribution of childcare affordability burden across U.S. states. The burden is calculated as annual infant childcare cost divided by median household income. Darker colors represent states where childcare consumes a larger share of household income.
- ▶ The map highlights that affordability pressure is not evenly distributed across the country. Some states experience substantially higher childcare burdens relative to income levels, emphasizing how national averages can obscure regional differences.

Key Insight

- ▶ National averages often hide local affordability pressure.
- ▶ In several states, infant childcare costs consume a significant share of household income, creating financial barriers for families and potentially affecting workforce participation.
- ▶ Visualizing these patterns helps highlight where childcare affordability challenges are most severe.

DSC640 Ayden Trusty Milestone 5

March 8, 2026

1 Milestone 5 Visuals

1.1 DSC640 Ayden Trusty

===== */

Title: Milestone 5

Author(s): Nathan Yau & Cole Nussbaumer Knafllic (Textbooks)

Date: 8 March 2026

Modified By: Ayden Trusty

Description: This program is to demonstrate my understanding/practice of using Python to review datasets and plot visuals to tell a story.

```
[1]: import pandas as pd
import matplotlib.pyplot as plt
```

```
[2]: file_path = "nationaldatabaseofchildcareprices.xlsx"
df = pd.read_excel(file_path)
```

Chart 1. Trend over time

```
[3]: trend = df.groupby("StudyYear")["MCInfant"].mean().sort_index()

plt.figure(figsize=(10, 4))
plt.plot(trend.index, trend.values, marker="o")
plt.title("Average Weekly Infant Center-Based Childcare Price Over Time")
plt.xlabel("Year")
plt.ylabel("Weekly Price ($)")
plt.tight_layout()
plt.savefig("01_trend_infant_center.png", dpi=300)
plt.show()
```

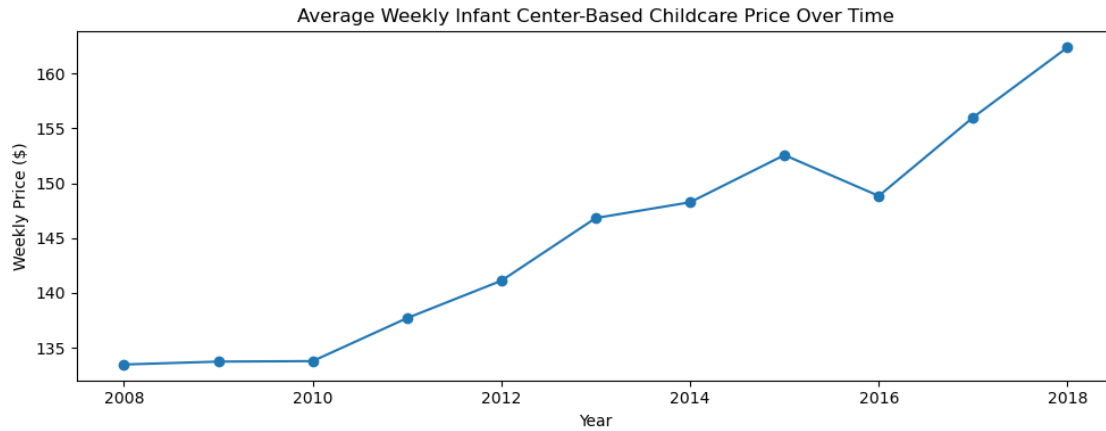
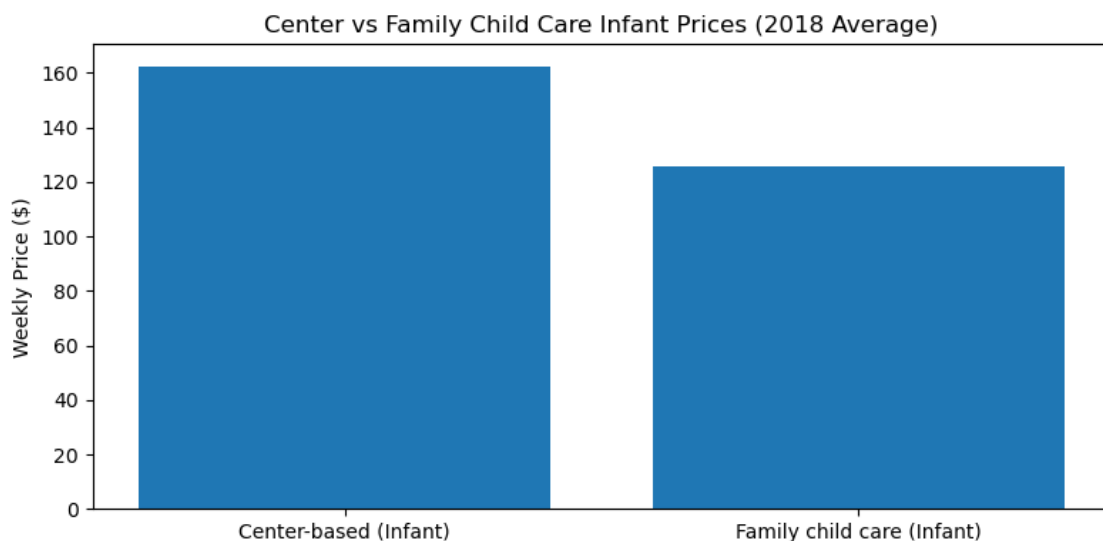


Chart 2. Care type comparison (2018)

```
[4]: df_2018 = df[df["StudyYear"] == 2018].copy()

center = df_2018["MCInfant"].mean()
fcc = df_2018["MFCCInfant"].mean()

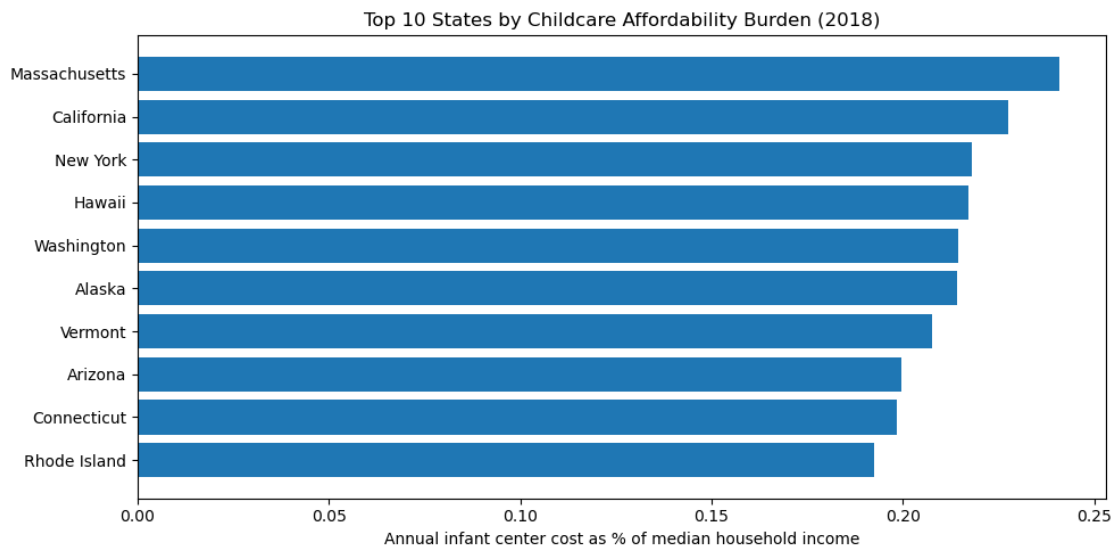
plt.figure(figsize=(8, 4))
plt.bar(["Center-based (Infant)", "Family child care (Infant)"], [center, fcc])
plt.title("Center vs Family Child Care Infant Prices (2018 Average)")
plt.ylabel("Weekly Price ($)")
plt.tight_layout()
plt.savefig("02_caretype_infant_2018.png", dpi=300)
plt.show()
```



```
[5]: df_2018 = df_2018[df_2018["MHI"].notna() & (df_2018["MHI"] > 0) &
      ↪df_2018["MCInfant"].notna()).copy()
df_2018["burden"] = (df_2018["MCInfant"] * 52) / df_2018["MHI"]

state_afford = (
    df_2018.groupby("State_Name")["burden"]
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

plt.figure(figsize=(10, 5))
plt.barh(state_afford.index[::-1], state_afford.values[::-1])
plt.title("Top 10 States by Childcare Affordability Burden (2018)")
plt.xlabel("Annual infant center cost as % of median household income")
plt.tight_layout()
plt.savefig("03_affordability_top10_states_2018.png", dpi=300)
plt.show()
```



```
[7]: import pandas as pd
import plotly.express as px

# 1) Load only needed columns since previous attempt was extremely slow
df = pd.read_excel(
    "nationaldatabaseofchildcareprices.xlsx",
    usecols=["StudyYear", "State_Abbreviation", "MCInfant", "MHI"]
)
```

```

# 2) Keep 2018 only
df_2018 = df[df["StudyYear"] == 2018].copy()

# 3) Create affordability burden
df_2018["burden"] = (df_2018["MCInfant"] * 52) / df_2018["MHI"]

# 4) Average burden by state
state_burden = (
    df_2018.groupby("State_Abbreviation")["burden"]
    .mean()
    .reset_index()
)

# 5) Make the US heatmap
fig = px.choropleth(
    state_burden,
    locations="State_Abbreviation",
    locationmode="USA-states",
    color="burden",
    scope="usa",
    labels={"burden": "Affordability Burden"},
    title="Childcare Affordability Burden by State (2018)"
)

fig.show()

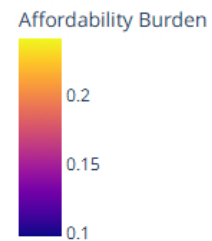
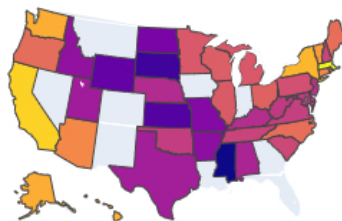
```

[]:

```
# 5) Make the US heatmap
fig = px.choropleth(
    state_burden,
    locations="State_Abbreviation",
    locationmode="USA-states",
    color="burden",
    scope="usa",
    labels={"burden": "Affordability Burden"},
    title="Childcare Affordability Burden by State (2018)"
)

fig.show()
```

Childcare Affordability Burden by State (2018)



[]:

